Palm Oil Retention & Melting System

Submerged Hydronic Coil Design — Technical Proposal

Based on the *VCH Sizing Framework, Second Edition*
A Unified Design Methodology for Submerged Hydronic Coil Systems

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This document presents thermal sizing, heat loss analysis, NTU-ε performance evaluation, process control architecture, and production capacity validation for a $3 \times 10\,\mathrm{kL}$ palm oil storage and $1 \times 5\,\mathrm{kL}$ replenishment tank system, integrated with a passive solar thermal loop and auxiliary electric immersion heating.

Contents

Executive Summary

This document presents the thermal design and sizing of a submerged hydronic coil system for the retention, melting, and trim-heating of palm oil at a Kathmandu processing facility, developed as part of an internal engineering study at Mechatronics Engineering Pvt. Ltd. (MEPL). The system comprises three 10 kL retention tanks ($T_{10,A\text{–}C}$) that receive and melt solid palm oil, and a single 5 kL insulated replenishment tank (T_5) that trim-heats the molten oil to the 60 °C process target. Heat is supplied primarily by a passive solar thermal loop, backed by a 3×12 kW electric immersion auxiliary battery for off-solar hours. The work presented here — covering vessel heat loss, coil and water-side hydraulics, two-phase U-value determination, NTU-ε performance, energy and melt-time budgeting, and process control architecture — was independently researched, modelled, and authored by Sambridda Ranabhat.

Headline Results

Metric	Result
Kathmandu annual solar reliability	75.9%
Single-tank capacity (primary, 75 °C)	**(Depends on Pipe configuration)**
Single-tank capacity (worst case, 70 °C)	**(Depends on Pipe configuration)**
Safety margin vs. 2 t/day target	2.42× – 3.63×
Effective capacity, 3 tanks staggered	≥ 20 t/day (10× margin)
Tank heat loss, melt phase (uninsulated)	≈ 1.07 kW
Overall U — solid / liquid phase	**(SEE Table 7)**
T_5 outlet temperature vs. target	**(Varies between 55 °C and 40 °C)**

Key Findings

- **The 2 t/day target is met with substantial margin** even under the worst-case 70 °C thermal driver scenario, using a single uninsulated T_{10} tank. With all three tanks staggered, the system delivers a 10× margin over target.
- **The solid-phase heat transfer coefficient is the system bottleneck**, governing the ≈28–33 hr melt time per tank. Mechanical agitation or a higher driver temperature are identified as the primary levers if future throughput targets increase.
- **T_5 is thermally over-specified** for its sensible-heating duty ($NTU \approx 20$), so a duty-cycle loop is required to hold the Dynamic Upper Thermal Boundary of $T_f(x, y, m_p) >$ 55 °C based on dynamic set point rather than static set point preventing overshooting above 60 °C.
- **The control system is fully demand-driven**: T_5 is the process master, T_{10} tanks are ranked continuously by a volume/temperature-weighted scoring system, and oil distribution to downstream lines is priority-queued and HMI-configurable.
- **Insulating T_{10}** is recommended as a low-cost efficiency gain, reducing hold-phase losses by ≈ 85%.

The remainder of this document develops each of these results from first principles, beginning with the Kathmandu climate baseline that underpins every thermal calculation that follows.

1 System Overview

1.1 Scope and Purpose

This proposal sizes the hydronic heating coil system for a palm oil storage and batch processing facility in Kathmandu, Nepal. The system must:

1. Receive solid palm oil (stored at ambient $\approx 18.1\,°\mathrm{C}$) into three 10 kL retention tanks ($\mathrm{T}_{10,A}$, $\mathrm{T}_{10,B}$, $\mathrm{T}_{10,C}$), melt it, and hold it in a liquid state.
2. Transfer liquid palm oil via a high-pressure pump (HPP) to a single 5 kL insulated replenishment tank (T_5), where it is trim-heated to the process target of $60\,°\mathrm{C}$.
3. Achieve a daily throughput target of ≥ 2 tonnes of processed liquid palm oil.
4. Derive primary heat from a passive solar thermal loop (5× evacuated-tube collectors, 2500 L reservoir) with an auxiliary thermal battery of 3×12 kW SS 316L immersion elements for off-solar hours.

1.2 Vessel Summary

Table 1: Vessel inventory and key parameters.

Tag	Role	Volume	Dia. (m)	Cyl. Ht. (m)	Bottom	Insulated
T_{10} (×3)	Long-term storage / melting	10 kL	2.32	2.20	Flat	No
T_5	Replenishment / trim-heat	5 kL	1.80	2.00	Conical	Yes

1.3 Thermal Driver

The primary thermal driver is the solar loop reservoir, which delivers hot water at a weighted annual average of $\approx 76\,°\mathrm{C}$–$78\,°\mathrm{C}$ based on the Kathmandu seasonal irradiance analysis. For this sizing, the following design temperatures are used:

- **Primary case:** $T_{\mathrm{hot}} = 75\,°\mathrm{C}$
- **Worst case:** $T_{\mathrm{hot}} = 70\,°\mathrm{C}$

The paper calculates for the delivery medium to be hot water supplied via 1.5 inch CPVC pipe or 1 inch CPVC pipe from the thermal battery to the tank coil headers.It also accounts for different coil sizes and computes for A.A, A.B, B.A and B.B.

2 Kathmandu Climate Baseline and Solar Thermal Achievability

2.1 Weighted Average of Usable Solar Days

To determine the long-term reliability of the solar utility system in Kathmandu, seasonal irradiance data is aggregated into a single annual usability metric. The weighted average is computed by dividing the total usable solar days by the total days in the year:

$$\overline{U} = \frac{\sum_i (\text{Days}_i \times \text{Usable\%}_i)}{\sum_i \text{Days}_i} = \frac{\text{Total Usable Days}}{\text{Total Days}} \tag{1}$$

Substituting the seasonal station metrics:

$$\text{Total Usable Days} = 73\ (\text{Winter}) + 82\ (\text{Spring}) + 67\ (\text{Monsoon}) + 55\ (\text{Autumn}) = 277 \text{ days}$$

$$\text{Total Days} = 90 + 92 + 122 + 61 = 365 \text{ days}$$

$$\overline{U} = \frac{277}{365} \approx 0.7589 \implies \mathbf{75.9}\% \text{ annual solar reliability} \tag{2}$$

2.2 Thermal Achievability of the 75 °C Loop Target

To confirm that the utility loop can reach and sustain 75 °C without solar input, a dead-start worst case is evaluated. The auxiliary system consists of 3 × 12 kW SS 316L immersion elements ($P_{\text{aux}} = 36$ kW total) acting through an SSR-modulated controller on a 300 L insulated active buffer volume. The initial temperature is taken as ambient: $T_{\text{init}} = 24$ °C.

Total Thermal Energy Requirement

$$Q = m\, C_p\, \Delta T = 300 \text{ kg} \times 4184 \text{ J kg}^{-1}\text{K}^{-1} \times (75 - 24) \text{ K} = 64.0\,\text{MJ} \tag{3}$$

Time-to-Temperature Response

$$t = \frac{Q}{P_{\text{aux}}} = \frac{64{,}015{,}200 \text{ J}}{36{,}000 \text{ W}} \approx 1778\,\text{s} \approx 29.6\,\text{min} \tag{4}$$

The auxiliary electric system can drive the entire active buffer from ambient to 75 °C in under 30 minutes from a cold start, confirming that maintaining a steady-state 75 °C process loop is operationally stable and achievable even during zero-solar periods.

2.3 Weighted Average Ambient Temperature — Kathmandu

The ambient design temperature used throughout this proposal ($T_\infty = 18.1\,°\mathrm{C}$) is derived from the Kathmandu historical monthly average, weighted by the number of days in each month.

$$\bar{T}_\infty = \frac{\sum_{i=1}^{12}(x_i \cdot w_i)}{\sum_{i=1}^{12} w_i} \tag{5}$$

where x_i is the monthly average temperature and w_i the number of days in month i.

Table 2: Kathmandu monthly temperature data and weighted contribution.

Month	**Avg Temp x_i (°C)**	**Days w_i**	**Weighted $x_i w_i$**
January	9.4	31	291.4
February	12.2	28	341.6
March	16.1	31	499.1
April	20.0	30	600.0
May	22.2	31	688.2
June	23.9	30	717.0
July	23.3	31	722.3
August	23.3	31	722.3
September	22.2	30	666.0
October	19.4	31	601.4
November	14.4	30	432.0
December	10.6	31	328.6
Total		**365**	**6609.9**

$$\bar{T}_\infty = \frac{6609.9}{365} = 18.109\ldots \approx 18.1\,°\mathrm{C} \tag{6}$$

This value is used as T_∞ in all tank heat loss and initial oil temperature calculations throughout this document.

3 Vessel Geometry

3.1 T_{10} (10 kL Retention Tank)

Tank T_{10} is a mild-steel vessel with a cylindrical body and a flat bottom. The external surface area governs the heat loss calculation.

$$r_{10} = \frac{D_{10}}{2} = \frac{2.32}{2} = 1.16\,\mathrm{m} \tag{7}$$

$$A_{\mathrm{cyl}} = \pi D_{10} H_{10} = \pi \times 2.32 \times 2.20 = 16.035\,\mathrm{m}^2 \tag{8}$$

$$A_{\mathrm{base}} = \pi r_{10}^2 = \pi \times 1.16^2 = 4.229\,\mathrm{m}^2 \tag{9}$$

$$A_{\mathrm{vessel},10} = A_{\mathrm{cyl}} + A_{\mathrm{base}} = 16.035 + 4.229 = 20.264\,\mathrm{m}^2 \tag{10}$$

3.2 T_5 (5 kL Replenishment Tank)

Tank T_5 is a mild-steel vessel with a cylindrical body and a conical bottom (half-angle 30).

$$r_5 = 0.90\,\mathrm{m} \tag{11}$$

$$h_{\mathrm{cone},5} = r_5 \tan(30) = 0.90 \times 0.5774 = 0.520\,\mathrm{m} \tag{12}$$

$$\ell_{\mathrm{slant},5} = \sqrt{r_5^2 + h_{\mathrm{cone},5}^2} = \sqrt{0.90^2 + 0.520^2} = 1.040\,\mathrm{m} \tag{13}$$

$$A_{\mathrm{cyl},5} = \pi D_5 H_5 = \pi \times 1.80 \times 2.00 = 11.310\,\mathrm{m}^2 \tag{14}$$

$$A_{\mathrm{cone},5} = \pi r_5 \,\ell_{\mathrm{slant},5} = \pi \times 0.90 \times 1.040 = 2.940\,\mathrm{m}^2 \tag{15}$$

$$A_{\mathrm{vessel},5} = A_{\mathrm{cyl},5} + A_{\mathrm{cone},5} = 11.310 + 2.940 = 14.250\,\mathrm{m}^2 \tag{16}$$

4 Tank Heat Loss Analysis

4.1 Methodology

T_{10} is uninsulated mild steel ($t_{\mathrm{wall}} = 6\,\mathrm{mm}$, $k_{\mathrm{steel}} = 50\,\mathrm{W\,m^{-1}\,K^{-1}}$). Heat is lost by: (1) inner-surface convection from oil to wall, (2) conduction through the steel, (3) natural convection from the outer wall to ambient air. Radiation is excluded per design brief. The analysis is performed at the *melting phase* condition ($T_{\mathrm{oil}} = 36\,°\mathrm{C}$), which yields the minimum loss rate — a conservative lower bound that is subtracted from the coil's useful output.

Ambient temperature: $T_\infty = 18.1\,°\mathrm{C}$ (Kathmandu weighted annual average).

4.2 Conduction Resistance

$$R_{\mathrm{cond}} = \frac{t_{\mathrm{wall}}}{k_{\mathrm{steel}}} = \frac{0.006}{50} = 1.20 \times 10^{-4}\,\mathrm{m^2\,K\,W^{-1}} \tag{17}$$

4.3 External Natural Convection — Cylindrical Wall

Air properties evaluated at film temperature $T_f = (36 + 18.1)/2 = 27.1\,°\mathrm{C}$: $\mu_{\mathrm{air}} = 1.849 \times 10^{-5}\,\mathrm{Pa\,s}$, $k_{\mathrm{air}} = 0.026\,24\,\mathrm{W\,m^{-1}\,K^{-1}}$, $\mathrm{Pr}_{\mathrm{air}} = 0.7296$, $\nu_{\mathrm{air}} = 1.571 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$, $\beta_{\mathrm{air}} = 1/T_f(\mathrm{K}) = 3.322 \times 10^{-3}\,\mathrm{K^{-1}}$.

Rayleigh number (characteristic length $L = H_{10} = 2.20\,\mathrm{m}$):

$$Ra_L = \frac{g\,\beta_{\text{air}}\,\Delta T_{\text{vessel}}\,L^3}{\nu_{\text{air}}^2}\,\Pr_{\text{air}} = \frac{9.81 \times 3.322 \times 10^{-3} \times 17.9 \times 2.20^3}{(1.571 \times 10^{-5})^2} \times 0.7296 = 1.841 \times 10^{10} \tag{18}$$

Churchill–Chu correlation for a vertical surface:

$$\overline{Nu}_L = \left[0.825 + \frac{0.387\,Ra_L^{1/6}}{[1+(0.492/\Pr)^{9/16}]^{8/27}}\right]^2 = \left[0.825 + \frac{0.387 \times (1.841 \times 10^{10})^{1/6}}{1.102}\right]^2 = 307.3 \tag{19}$$

$$h_{\text{ext,cyl}} = \frac{307.3 \times 0.02624}{2.20} = 3.665\,\mathrm{W\,m^{-2}\,K^{-1}} \tag{20}$$

4.4 External Natural Convection — Flat Base

The flat circular base is warmer than ambient and faces downward (hot surface facing down), so buoyancy suppresses convection. The Morgan–McAdams correlation for a heated horizontal plate facing downward applies, with characteristic length $L_c = A_{\text{base}}/P_{\text{base}} = r_{10}/2 = 0.58\,\mathrm{m}$:

$$Ra_{L_c} = \frac{g\,\beta_{\text{air}}\,\Delta T_{\text{vessel}}\,L_c^3}{\nu_{\text{air}}^2}\,\Pr_{\text{air}} = \frac{9.81 \times 3.322 \times 10^{-3} \times 17.9 \times 0.58^3}{(1.571 \times 10^{-5})^2} \times 0.7296 = 3.374 \times 10^{8} \tag{21}$$

$$\overline{Nu}_{L_c} = 0.27\,Ra_{L_c}^{0.25} = 0.27 \times (3.374 \times 10^8)^{0.25} = 36.59 \;\Rightarrow\; h_{\text{ext,base}} = \frac{36.59 \times 0.02624}{0.58} = 1.656\,\mathrm{W\,m^{-2}\,K^{-1}} \tag{22}$$

4.5 Overall Heat Loss Coefficients and $\dot{Q}_{\text{loss}}$

Inner surface convection during solid/melting phase: $h_{\text{in,vessel}} = 35\,\mathrm{W\,m^{-2}\,K^{-1}}$ (conduction-dominated, stagnant melt sleeve — see Section 7).

$$U_{\text{ves,cyl}} = \left(\frac{1}{35} + 1.20 \times 10^{-4} + \frac{1}{3.665}\right)^{-1} = 3.317\,\mathrm{W\,m^{-2}\,K^{-1}} \tag{23}$$

$$U_{\text{ves,base}} = \left(\frac{1}{35} + 1.20 \times 10^{-4} + \frac{1}{1.656}\right)^{-1} = 1.581\,\mathrm{W\,m^{-2}\,K^{-1}} \tag{24}$$

$$\dot{Q}_{\text{loss,cyl}} = U_{\text{ves,cyl}} \times A_{\text{cyl}} \times \Delta T = 3.317 \times 16.035 \times 17.9 = 951.9\,\mathrm{W} \tag{25}$$

$$\dot{Q}_{\text{loss,base}} = U_{\text{ves,base}} \times A_{\text{base}} \times \Delta T = 1.581 \times 4.229 \times 17.9 = 119.7\,\mathrm{W} \tag{26}$$

$$\boxed{\dot{Q}_{\text{loss,total}} = 951.9 + 119.7 = 1071.6\,\text{W} \approx 1.07\,\text{kW}} \tag{27}$$

Note: This loss figure applies at the melting-phase condition ($T_{\text{oil}} = 36\,°\text{C}$). During the subsequent liquid-phase hold at 60 °C, the loss rises to approximately 2.6 kW, which must be met by the auxiliary immersion elements or solar loop during holding periods.

5 Coil Configuration

5.1 Material and Dimensions

Two coil bore options are evaluated: the baseline **1.5-inch NPS** (Coil A) and the alternative **1-inch NPS** (Coil B), both in SS 304 Schedule 10S.

Table 3: Coil pipe dimensions and thermal conductivity.

Parameter	Symbol	Coil A (1.5-inch)	Coil B (1-inch)
Outer diameter	$d_{o,\text{coil}}$	48.26 mm	33.40 mm
Inner diameter	$d_{i,\text{coil}}$	42.72 mm	27.86 mm
Wall thickness	t_c	2.77 mm	2.77 mm
d_o/d_i	—	1.1297	1.1989
Thermal conductivity (SS 304)	k_{SS}	$16.3\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$	

Note: Both coil sizes share the same Schedule 10S wall thickness of 2.77 mm. The 1-inch coil Sch 10S inner diameter (27.86 mm) differs from the 1-inch *supply pipe* Sch 40 CPVC inner diameter (26.64 mm); these are separate components.

5.2 Helix Geometry

Hot water is fed from the *bottom* of the helix (counter-flow to the rising natural convection currents of the oil), maximising the temperature driving force along the entire coil length.

The tank geometry fixes the helix diameter. Both coil sizes are accommodated within the same $D_{\text{helix}} = 2.072$ m constraint. The practical coil length is held constant at the VCH-optimised value for each tank; only the outer surface area changes with the coil bore.

Table 4: Coil geometric parameters for T_{10} and T_5.

		T_{10} (10 kL)		T_5 (5 kL)	
Parameter	Symbol	Coil A	Coil B	Coil A	Coil B
Helix diameter (m)	D_{helix}	2.072		1.711	
Turns	N	21	—	19	—
Practical length (m)	L_{prac}	90.0		65.0	
Coil surface area (m^2)	A_{coil}	13.68	9.44	9.88	6.82

The 1-inch coil surface areas are derived as $A = \pi\, d_{o,\text{coil}}\, L_\text{prac}$:

$$A_\text{coil,B,T10} = \pi \times 0.03340 \times 90.0 = 9.44\,\text{m}^2 \tag{28}$$

$$A_\text{coil,B,T5} = \pi \times 0.03340 \times 65.0 = 6.82\,\text{m}^2 \tag{29}$$

5.3 Wall Conduction Resistance of Coil

$$R_\text{wall,A} = \frac{d_{o,A}\ln(d_{o,A}/d_{i,A})}{2\,k_\text{SS}} = \frac{0.04826 \times \ln(1.1297)}{2 \times 16.3} = 1.81 \times 10^{-4}\,\text{m}^2\,\text{K}\,\text{W}^{-1} \tag{30}$$

$$R_\text{wall,B} = \frac{d_{o,B}\ln(d_{o,B}/d_{i,B})}{2\,k_\text{SS}} = \frac{0.03340 \times \ln(1.1989)}{2 \times 16.3} = 1.86 \times 10^{-4}\,\text{m}^2\,\text{K}\,\text{W}^{-1} \tag{31}$$

6 Water-Side Hydraulics and Convection

Four configurations are evaluated by combining two supply pipe sizes (prefix letter) with two coil bore sizes (suffix letter):

Case	Supply pipe	Coil pipe
A.A	1.5-inch NPS Sch 40 CPVC	1.5-inch NPS Sch 10S SS 304
A.B	1.5-inch NPS Sch 40 CPVC	1-inch NPS Sch 10S SS 304
B.A	1-inch NPS Sch 40 CPVC	1.5-inch NPS Sch 10S SS 304
B.B	1-inch NPS Sch 40 CPVC	1-inch NPS Sch 10S SS 304

Supply pipe dimensions are unchanged from the original analysis (Table ??). Water properties at $T_\text{hot} = 75\,°\text{C}$ are also unchanged: $\rho_w = 974.9\,\text{kg}\,\text{m}^{-3}$, $C_{p,w} = 4193\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$, $\mu_w = 3.78 \times 10^{-4}\,\text{Pa}\,\text{s}$, $k_w = 0.6636\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$, $\text{Pr}_w = 2.39$.

6.1 Flow Rate

The maximum supply velocity $V_\text{max} = 1.5\,\text{m}\,\text{s}^{-1}$ applies at the supply pipe bore. Cases A.A and A.B share the same supply pipe (1.5-inch), so the mass flow rate is identical for both; likewise for Cases B.A and B.B (1-inch supply).

$$\dot{m}_{w,\text{Sup-A}} = \rho_w \frac{\pi d_{i,\text{sup},A}^2}{4} V_\text{max} = 974.9 \times 1.433 \times 10^{-3} \times 1.5 = 2.096\,\text{kg}\,\text{s}^{-1} \tag{32}$$

$$\dot{m}_{w,\text{Sup-B}} = \rho_w \frac{\pi d_{i,\text{sup},B}^2}{4} V_\text{max} = 974.9 \times 5.574 \times 10^{-4} \times 1.5 = 0.815\,\text{kg}\,\text{s}^{-1} \tag{33}$$

6.2 Coil-Side Velocity and Reynolds Number

Inside the helical coil the actual flow velocity depends on the *coil* bore, not the supply pipe. The cross-sectional areas are:

$$A_{i,\text{coil-A}} = \frac{\pi \times 0.04272^2}{4} = 1.433 \times 10^{-3}\,\text{m}^2 \tag{34}$$

$$A_{i,\text{coil-B}} = \frac{\pi \times 0.02786^2}{4} = 6.094 \times 10^{-4}\,\text{m}^2 \tag{35}$$

The in-coil velocities and Reynolds numbers for each of the four sub-cases follow directly:

Table 5: Coil-side flow conditions for all four sub-cases.

Case	$\dot{m}_w$ (kg/s)	V_{coil} (m/s)	Re_{coil}	De_{coil}	Regime
A.A	2.096	1.500	165,269	23,731	turbulent
A.B	2.096	3.527	253,420	29,386	turbulent
B.A	0.815	0.583	64,268	9,228	turbulent
B.B	0.815	1.372	98,548	11,427	turbulent

Note: In Case A.B the 1.5-inch supply delivers $2.096\,\text{kg}\,\text{s}^{-1}$ through a 1-inch coil bore, raising the in-coil velocity to $3.527\,\text{m}\,\text{s}^{-1}$ — well above the supply-pipe design limit, but acceptable for the coil itself (SS 304 tubing tolerates higher velocities without erosion at these flow rates). This high velocity is responsible for the elevated h_i in Case A.B but does not benefit overall U because the outer oil-side resistance dominates.

6.3 Inner Heat Transfer Coefficient

Gnielinski correlation with Kubair–Kuloor helical enhancement, applied at the coil-side Reynolds number. All four sub-cases share $D_{\text{helix}} = 2.072\,\text{m}$.

$$f = \big(0.79 \ln Re_{\text{coil}} - 1.64\big)^{-2} \tag{36}$$

$$Nu_{\text{str}} = \frac{(f/8)(Re_{\text{coil}} - 1000)\,\text{Pr}_w}{1 + 12.7\sqrt{f/8}(\text{Pr}_w^{2/3} - 1)} \tag{37}$$

$$Nu_{\text{helix}} = Nu_{\text{str}} \left(1 + 3.54\,\frac{d_{i,\text{coil}}}{D_{\text{helix}}}\right) \tag{38}$$

$$h_{i,w} = \frac{Nu_{\text{helix}}\,k_w}{d_{i,\text{coil}}} \tag{39}$$

Case A.A — 1.5-inch supply / 1.5-inch coil

$$f_{A.A} = (0.79 \ln 165{,}269 - 1.64)^{-2} = 1.622 \times 10^{-2} \tag{40}$$

$$Nu_{\text{str,A.A}} = 548.80 \tag{41}$$

$$Nu_{\text{helix,A.A}} = 548.80 \times \left(1 + 3.54 \times \frac{0.04272}{2.072}\right) = 588.85 \tag{42}$$

$$h_{i,A.A} = \frac{588.85 \times 0.6636}{0.04272} = 9147\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1} \tag{43}$$

Case A.B — 1.5-inch supply / 1-inch coil

$$f_{A.B} = (0.79 \ln 253{,}420 - 1.64)^{-2} = 1.491 \times 10^{-2} \tag{44}$$

$$Nu_{\text{str,A.B}} = 785.24 \tag{45}$$

$$Nu_{\text{helix,A.B}} = 785.24 \times \left(1 + 3.54 \times \frac{0.02786}{2.072}\right) = 822.62 \tag{46}$$

$$h_{i,A.B} = \frac{822.62 \times 0.6636}{0.02786} = 19.594\,\mathrm{W\,m^{-2}\,K^{-1}} \tag{47}$$

Case B.A — 1-inch supply / 1.5-inch coil

$$f_{B.A} = (0.79 \ln 64{,}268 - 1.64)^{-2} = 1.980 \times 10^{-2} \tag{48}$$

$$Nu_{\text{str,B.A}} = 249.94 \tag{49}$$

$$Nu_{\text{helix,B.A}} = 249.94 \times \left(1 + 3.54 \times \frac{0.04272}{2.072}\right) = 268.18 \tag{50}$$

$$h_{i,B.A} = \frac{268.18 \times 0.6636}{0.04272} = 4166\,\mathrm{W\,m^{-2}\,K^{-1}} \tag{51}$$

Case B.B — 1-inch supply / 1-inch coil

$$f_{B.B} = (0.79 \ln 98{,}548 - 1.64)^{-2} = 1.805 \times 10^{-2} \tag{52}$$

$$Nu_{\text{str,B.B}} = 356.56 \tag{53}$$

$$Nu_{\text{helix,B.B}} = 356.56 \times \left(1 + 3.54 \times \frac{0.02786}{2.072}\right) = 373.54 \tag{54}$$

$$h_{i,B.B} = \frac{373.54 \times 0.6636}{0.02786} = 8897\,\mathrm{W\,m^{-2}\,K^{-1}} \tag{55}$$

Note — inner coefficient vs. overall U: Case A.B achieves the highest inner coefficient ($h_i = 19.594\,\mathrm{W\,m^{-2}\,K^{-1}}$) because the large supply flow is squeezed through a narrow coil bore, roughly doubling the wall velocity gradient relative to Case A.A. Case B.A has the lowest ($h_i = 4166\,\mathrm{W\,m^{-2}\,K^{-1}}$) because the small supply flow is spread across the full 1.5-inch bore. However, as Section 7 shows, the outer oil-side resistance ($1/h_o$) dominates $1/U$ by a factor of $\geq 50\times$ in the solid phase and $\geq 8\times$ in the liquid phase, so variations in h_i across all four cases have negligible impact on U_{solid} and minor impact on U_{liquid}.

7 Two-Phase U-Value Determination

7.1 Fouling Allowances

Table 6: Fouling resistances applied per TEMA standards.

Surface	Condition	R_f ($m^2{\cdot}K{\cdot}W^{-1}$)
Inner (water side)	Soft water, low cycling	1.0×10^{-4}
Outer (palm oil side)	Palm oil at 60 °C, convection	9.0×10^{-4}

7.2 Wall Conduction Resistance

Already established in Section 1.3: $R_{\text{wall,coil-A}} = 1.81 \times 10^{-4}\,\mathrm{m^2\,K\,W^{-1}}$ and $R_{\text{wall,coil-B}} = 1.86 \times 10^{-4}\,\mathrm{m^2\,K\,W^{-1}}$.

7.3 Overall Heat Transfer Coefficient

The general resistance equation (referred to the outer surface area) is:

$$\frac{1}{U} = \frac{1}{h_{i,w}} \cdot \frac{d_o}{d_i} + R_{f,i} \cdot \frac{d_o}{d_i} + R_{\text{wall}} + R_{f,o} + \frac{1}{h_o} \tag{56}$$

where $h_o = 35\,\mathrm{W\,m^{-2}\,K^{-1}}$ (solid phase) and $h_o = 138.9\,\mathrm{W\,m^{-2}\,K^{-1}}$ (liquid phase, identical for all sub-cases since it is governed by tank geometry, not pipe size).

Case A.A

$$\frac{1}{U_{\text{solid,A.A}}} = \frac{1.1297}{9147} + 1.1297 \times 10^{-4} + 1.81 \times 10^{-4} + 9.0 \times 10^{-4} + \frac{1}{35} = 2.989 \times 10^{-2}$$
$$\Rightarrow \boxed{U_{\text{solid,A.A}} = 33.46\,\mathrm{W\,m^{-2}\,K^{-1}}} \tag{57}$$

$$\frac{1}{U_{\text{liquid,A.A}}} = \frac{1.1297}{9147} + 1.1297 \times 10^{-4} + 1.81 \times 10^{-4} + 9.0 \times 10^{-4} + \frac{1}{138.9} = 8.516 \times 10^{-3}$$
$$\Rightarrow \boxed{U_{\text{liquid,A.A}} = 117.4\,\mathrm{W\,m^{-2}\,K^{-1}}} \tag{58}$$

Case A.B

$$\frac{1}{U_{\text{solid,A.B}}} = \frac{1.1989}{19{,}594} + 1.1989 \times 10^{-4} + 1.86 \times 10^{-4} + 9.0 \times 10^{-4} + \frac{1}{35} = 2.984 \times 10^{-2}$$
$$\Rightarrow \boxed{U_{\text{solid,A.B}} = 33.51\,\mathrm{W\,m^{-2}\,K^{-1}}} \tag{59}$$

$$\frac{1}{U_{\text{liquid,A.B}}} = \frac{1.1989}{19{,}594} + 1.1989 \times 10^{-4} + 1.86 \times 10^{-4} + 9.0 \times 10^{-4} + \frac{1}{138.9} = 8.466 \times 10^{-3}$$
$$\Rightarrow \boxed{U_{\text{liquid,A.B}} = 118.1\,\mathrm{W\,m^{-2}\,K^{-1}}} \tag{60}$$

Case B.A

$$\frac{1}{U_{\text{solid,B.A}}} = \frac{1.1297}{4166} + 1.1297 \times 10^{-4} + 1.81 \times 10^{-4} + 9.0 \times 10^{-4} + \frac{1}{35} = 3.004 \times 10^{-2}$$

$$\Rightarrow \boxed{U_{\text{solid,B.A}} = 33.29\,\mathrm{W\,m^{-2}\,K^{-1}}} \tag{61}$$

$$\frac{1}{U_{\text{liquid,B.A}}} = \frac{1.1297}{4166} + 1.1297 \times 10^{-4} + 1.81 \times 10^{-4} + 9.0 \times 10^{-4} + \frac{1}{138.9} = 8.664 \times 10^{-3}$$

$$\Rightarrow \boxed{U_{\text{liquid,B.A}} = 115.4\,\mathrm{W\,m^{-2}\,K^{-1}}} \tag{62}$$

Case B.B

$$\frac{1}{U_{\text{solid,B.B}}} = \frac{1.1989}{8897} + 1.1989 \times 10^{-4} + 1.86 \times 10^{-4} + 9.0 \times 10^{-4} + \frac{1}{35} = 2.991 \times 10^{-2}$$

$$\Rightarrow \boxed{U_{\text{solid,B.B}} = 33.43\,\mathrm{W\,m^{-2}\,K^{-1}}} \tag{63}$$

$$\frac{1}{U_{\text{liquid,B.B}}} = \frac{1.1989}{8897} + 1.1989 \times 10^{-4} + 1.86 \times 10^{-4} + 9.0 \times 10^{-4} + \frac{1}{138.9} = 8.540 \times 10^{-3}$$

$$\Rightarrow \boxed{U_{\text{liquid,B.B}} = 117.1\,\mathrm{W\,m^{-2}\,K^{-1}}} \tag{64}$$

Table 7: Full resistance breakdown across all four sub-cases ($\times 10^{-3}\,\mathrm{m^2{\cdot}K{\cdot}W^{-1}}$). Phase 1 = solid; Phase 2 = liquid.

	Case A.A		Case A.B		Case B.A		Case B.B	
Resistance component	Ph 1	Ph 2	Ph 1	Ph 2	Ph 1	Ph 2	Ph 1	Ph 2
$1/(h_i \cdot d_o/d_i)$	0.124	0.124	0.061	0.061	0.271	0.271	0.135	0.135
$R_{f,i} \cdot (d_o/d_i)$	0.113	0.113	0.120	0.120	0.113	0.113	0.120	0.120
R_{wall}	0.181	0.181	0.186	0.186	0.181	0.181	0.186	0.186
$R_{f,o}$	0.900	0.900	0.900	0.900	0.900	0.900	0.900	0.900
$1/h_o$	28.571	7.199	28.571	7.199	28.571	7.199	28.571	7.199
Total $1/U$	29.888	8.516	29.838	8.466	30.036	8.664	29.912	8.540
U (W/m^{2}K)	**33.46**	**117.4**	**33.51**	**118.1**	**33.29**	**115.4**	**33.43**	**117.1**

The spread in U_{solid} across all four cases is only $\Delta U = 0.22\,\mathrm{W\,m^{-2}\,K^{-1}}$ (0.66%), and in U_{liquid} only $\Delta U = 2.7\,\mathrm{W\,m^{-2}\,K^{-1}}$ (2.3%). The outer oil-side resistance ($1/h_o$) accounts for >95% of $1/U_{\text{solid}}$ in every case, confirming that neither supply pipe size nor coil bore has a meaningful effect on the overall heat transfer coefficient. The performance differences between sub-cases arise almost entirely from differences in coil surface area (A_{coil}) and water-side capacity rate (C_w).

8 NTU-ε Thermal Performance — T$_{10}$ Melting Phase

8.1 Capacity Rates

During the melting phase, palm oil absorbs energy at constant temperature ($T_{\text{melt}} = 36\,°\mathrm{C}$), giving $C_r = 0$.

$$C_{w,\text{Sup-A}} = \dot{m}_{w,\text{Sup-A}}\, C_{p,w} = 2.096 \times 4193 = 8789\,\mathrm{W\,K^{-1}} \tag{65}$$

$$C_{w,\text{Sup-B}} = \dot{m}_{w,\text{Sup-B}}\, C_{p,w} = 0.815 \times 4193 = 3418\,\mathrm{W\,K^{-1}} \tag{66}$$

Cases A.A and A.B share $C_w = 8789\,\mathrm{W\,K^{-1}}$; Cases B.A and B.B share $C_w = 3418\,\mathrm{W\,K^{-1}}$.

8.2 Number of Transfer Units and Effectiveness

For $C_r = 0$: $\varepsilon = 1 - e^{-NTU}$.

$$NTU = \frac{U_{\text{solid}}\, A_{\text{coil}}}{C_w} \tag{67}$$

Case A.A

$$NTU_{\text{solid,A.A}} = \frac{33.46 \times 13.68}{8789} = 0.05195 \qquad \varepsilon_{\text{solid,A.A}} = 1 - e^{-0.05195} = 0.05062 \tag{68}$$

Case A.B

$$NTU_{\text{solid,A.B}} = \frac{33.51 \times 9.44}{8789} = 0.03601 \qquad \varepsilon_{\text{solid,A.B}} = 1 - e^{-0.03601} = 0.03537 \tag{69}$$

Case B.A

$$NTU_{\text{solid,B.A}} = \frac{33.29 \times 13.68}{3418} = 0.13292 \qquad \varepsilon_{\text{solid,B.A}} = 1 - e^{-0.13292} = 0.12447 \tag{70}$$

Case B.B

$$NTU_{\text{solid,B.B}} = \frac{33.43 \times 9.44}{3418} = 0.09238 \qquad \varepsilon_{\text{solid,B.B}} = 1 - e^{-0.09238} = 0.08824 \tag{71}$$

8.3 Actual and Net Heat Transfer at T_{10}

$$\dot{Q}_{\text{actual}} = \varepsilon\, C_w\, (T_{\text{hot}} - T_{\text{melt}}) \tag{72}$$

$$\dot{Q}_{\text{net}} = \dot{Q}_{\text{actual}} - \dot{Q}_{\text{loss}} \qquad (\dot{Q}_{\text{loss}} = 1071.6\,\text{W}) \tag{73}$$

Case A.A

$$\dot{Q}_{\text{net},75,A.A} = 0.05062 \times 8789 \times 39 - 1072 = 16.279\,\text{W} \approx 16.28\,\text{kW} \tag{74}$$

$$\dot{Q}_{\text{net},70,A.A} = 0.05062 \times 8789 \times 34 - 1072 = 14.054\,\text{W} \approx 14.05\,\text{kW} \tag{75}$$

Water outlet temperatures: $T_{w,\text{out}} = 73.0\,°\text{C}$ ($\Delta T_w = 2.0\,\text{K}$) at 75°C drive.

Case A.B

$$\dot{Q}_{\text{net},75,A.B} = 0.03537 \times 8789 \times 39 - 1072 = 11.052\,\text{W} \approx 11.05\,\text{kW} \tag{76}$$

$$\dot{Q}_{\text{net},70,A.B} = 0.03537 \times 8789 \times 34 - 1072 = 9.498\,\text{W} \approx 9.50\,\text{kW} \tag{77}$$

Water outlet temperatures: $T_{w,\text{out}} = 73.6\,°\text{C}$ ($\Delta T_w = 1.4\,\text{K}$) at 75°C drive. The small coil surface area limits total heat extraction even though h_i is highest in this case.

Case B.A

$$\dot{Q}_{\text{net},75,B.A} = 0.12447 \times 3418 \times 39 - 1072 = 15.519\,\text{W} \approx 15.52\,\text{kW} \tag{78}$$

$$\dot{Q}_{\text{net},70,B.A} = 0.12447 \times 3418 \times 34 - 1072 = 13.392\,\text{W} \approx 13.39\,\text{kW} \tag{79}$$

Water outlet temperatures: $T_{w,\text{out}} = 70.2\,°\text{C}$ ($\Delta T_w = 4.8\,\text{K}$) at 75°C drive.

Case B.B

$$\dot{Q}_{\text{net},75,B.B} = 0.08824 \times 3418 \times 39 - 1072 = 10.690\,\text{W} \approx 10.69\,\text{kW} \tag{80}$$

$$\dot{Q}_{\text{net},70,B.B} = 0.08824 \times 3418 \times 34 - 1072 = 9.182\,\text{W} \approx 9.18\,\text{kW} \tag{81}$$

Water outlet temperatures: $T_{w,\text{out}} = 71.6\,°\text{C}$ ($\Delta T_w = 3.4\,\text{K}$) at 75°C drive.

Design note — the dominant variable is coil surface area, not h_i. Comparing A.A (16.28 kW) with A.B (11.05 kW): both share the same supply flow and C_w, but Case A.B loses 5.2 kW net because the 1-inch coil presents only 9.44 m^2 versus the 1.5-inch coil's 13.68 m^2 — a 31% area reduction. The elevated $h_i = 19.594\,\mathrm{W\,m^{-2}\,K^{-1}}$ in Case A.B does not compensate, because $1/h_o$ remains the dominant resistance in both cases.

Comparing B.A (15.52 kW) with B.B (10.69 kW): the same surface-area penalty applies with the 1-inch supply. B.B is the worst performer across all four cases — smaller supply *and* smaller coil.

9 Energy Budget and Melt Time

9.1 Total Energy Requirement — Full 10 kL Tank

Unchanged from original analysis: $E_{\text{sensible}} = 318.6\,\mathrm{MJ}$, $E_{\text{latent}} = 1335.0\,\mathrm{MJ}$, $\mathbf{E}_{\text{total}} = 1653.6\,\mathrm{MJ}$.

Specific energy: $q = C_{p,s}\Delta T + L_f = 2000 \times 17.9 + 150{,}000 = 185.800\,\mathrm{J\,kg^{-1}}$.

9.2 Melt Time

$$t_{\text{melt}} = \frac{E_{\text{total}}}{\dot{Q}_{\text{net}}} \tag{82}$$

Case A.A

$$t_{\text{melt},75,A.A} = \frac{1653.6 \times 10^6}{16{,}279} = 101.581\,\mathrm{s} = 28.22\,\mathrm{hr} \tag{83}$$

$$t_{\text{melt},70,A.A} = \frac{1653.6 \times 10^6}{14{,}054} = 117.664\,\mathrm{s} = 32.68\,\mathrm{hr} \tag{84}$$

Case A.B

$$t_{\text{melt},75,A.B} = \frac{1653.6 \times 10^6}{11{,}052} = 149.622\,\mathrm{s} = 41.56\,\mathrm{hr} \tag{85}$$

$$t_{\text{melt},70,A.B} = \frac{1653.6 \times 10^6}{9{,}498} = 174.099\,\mathrm{s} = 48.36\,\mathrm{hr} \tag{86}$$

Case B.A

$$t_{\text{melt},75,B.A} = \frac{1653.6 \times 10^6}{15{,}519} = 106.554\,\mathrm{s} = 29.60\,\mathrm{hr} \tag{87}$$

$$t_{\text{melt},70,B.A} = \frac{1653.6 \times 10^6}{13{,}392} = 123.483\,\mathrm{s} = 34.30\,\mathrm{hr} \tag{88}$$

Case B.B

$$t_{\text{melt},75,B.B} = \frac{1653.6 \times 10^6}{10{,}690} = 154.687\,\text{s} = 42.97\,\text{hr} \tag{89}$$

$$t_{\text{melt},70,B.B} = \frac{1653.6 \times 10^6}{9{,}182} = 180.098\,\text{s} = 50.03\,\text{hr} \tag{90}$$

9.3 Instantaneous Melting Rate (at 75°C drive)

$$\dot{m}_{\text{melt}} = \frac{\dot{Q}_{\text{net}}}{q} \times 3600 \quad [\text{kg/hr}] \tag{91}$$

Table 8: Instantaneous melting rates across all four sub-cases.

Case	$\dot{Q}_{\text{net}}$ (kW)	$\dot{m}_{\text{melt}}$ (kg/hr)	$\dot{m}_{\text{melt}}$ (kg/min)
A.A	16.28	315.4	5.26
A.B	11.05	214.1	3.57
B.A	15.52	300.7	5.01
B.B	10.69	207.1	3.45

9.4 Solar Circuit Characterization — Kathmandu Conditions

Solar analysis is independent of supply pipe or coil size; all four sub-cases share the same collector performance.

Average operational irradiance: $G = 750\,\text{W}\,\text{m}^{-2}$, peak sun hours: $PSH = 4.61\,\text{hr}$, ambient temperature: $T_a = 18.1\,°\text{C}$, $\Delta T = T_f - T_a = 56.9\,°\text{C}$.

Flat Plate Collector (FPC)

$N = 15$ panels, $A_{c,\text{FPC}} = 22.674\,\text{m}^2$, $\eta_0 = 0.75$, $a_1 = 3.50\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$, $a_2 = 0.015\,\text{W}\,\text{m}^{-2}\,\text{K}^{-2}$.

$$\eta_{\text{FPC}} = 0.75 - \frac{3.50 \times 56.9}{750} - \frac{0.015 \times 56.9^2}{750} = 0.4197 = 41.97\,\% \tag{92}$$

$$\dot{Q}_{\text{peak,FPC}} = 7.14\,\text{kW} \qquad \mathbf{E}_{\text{solar,FPC}} = 118.45\,\text{MJ} \tag{93}$$

Evacuated Tube Collector (ETC)

$N = 5$ units, $A_{c,\text{ETC}} = 32.50\,\text{m}^2$, $\eta_0 = 0.70$, $a_1 = 1.50\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$, $a_2 = 0.005\,\text{W}\,\text{m}^{-2}\,\text{K}^{-2}$.

$$\eta_{\text{ETC}} = 0.70 - \frac{1.50 \times 56.9}{750} - \frac{0.005 \times 56.9^2}{750} = 0.5646 = 56.46\,\% \tag{94}$$

$$\dot{Q}_{\text{peak,ETC}} = 13.76\,\text{kW} \qquad \mathbf{E}_{\text{solar,ETC}} = 228.40\,\text{MJ} \tag{95}$$

9.5 Solar Integration — Full Batch and 2-Tonne Target

The solar yield is fixed; coverage fractions are therefore identical across all four sub-cases.

$$E_{\text{total,full}} = 1653.6\,\text{MJ} \tag{96}$$

$$E_{\text{total,2t}} = 2000 \times 185{,}800 \times 10^{-6} = 371.6\,\text{MJ} \tag{97}$$

		Batch coverage (%)		$E_{\text{auxiliary}}$ (MJ)	
Config.	E_{solar} (MJ)	Full batch	2-tonne	Full batch	2-tonne
FPC	118.45	7.16	31.88	1535.15	253.15
ETC	228.40	13.81	**61.46**	1425.20	**143.20**

The ETC configuration covers 61.46% of the 2 t/day minimum production energy demand from solar alone, reducing auxiliary consumption to 143.20 MJ — a 43.4% reduction versus the FPC auxiliary requirement. These figures apply equally to Cases A.A, A.B, B.A, and B.B.

9.6 Time to Process the 2-Tonne Daily Production Target

Rather than tracking whole-batch cycles against a 24-hour window, it is more operationally relevant to ask: *how long does a single T_{10} tank take to deliver the 2 t/day minimum order?*

The energy demand for 2000 kg is:

$$E_{2\text{t}} = 2000 \times q_{\text{specific}} = 2000 \times 185{,}800 = 371.6\,\text{MJ} \tag{98}$$

The melt time and total turnaround (including the 1.25 hr extraction and reload overhead) follow directly from the net heat delivery of each sub-case:

$$t_{\text{melt,2t}} = \frac{E_{2\text{t}}}{\dot{Q}_{\text{net}}} \qquad t_{\text{turn,2t}} = t_{\text{melt,2t}} + 1.25\,\text{hr} \tag{99}$$

Case A.A

$$t_{\text{melt,75,A.A}} = \frac{371.6 \times 10^6}{16{,}279} = 22.828\,\text{s} = 6.34\,\text{hr} \tag{100}$$

$$t_{\text{turn,75,A.A}} = 6.34 + 1.25 = 7.59\,\text{hr} \tag{101}$$

$$t_{\text{melt,70,A.A}} = \frac{371.6 \times 10^6}{14{,}054} = 26.438\,\text{s} = 7.34\,\text{hr} \tag{102}$$

$$t_{\text{turn,70,A.A}} = 7.34 + 1.25 = 8.59\,\text{hr} \tag{103}$$

Case A.B

$$t_{\text{melt},75,\text{A.B}} = \frac{371.6 \times 10^6}{11{,}052} = 33.627\,\text{s} = 9.34\,\text{hr} \tag{104}$$

$$t_{\text{turn},75,\text{A.B}} = 9.34 + 1.25 = 10.59\,\text{hr} \tag{105}$$

$$t_{\text{melt},70,\text{A.B}} = \frac{371.6 \times 10^6}{9{,}498} = 39.124\,\text{s} = 10.87\,\text{hr} \tag{106}$$

$$t_{\text{turn},70,\text{A.B}} = 10.87 + 1.25 = 12.12\,\text{hr} \tag{107}$$

Case B.A

$$t_{\text{melt},75,\text{B.A}} = \frac{371.6 \times 10^6}{15{,}519} = 23.944\,\text{s} = 6.65\,\text{hr} \tag{108}$$

$$t_{\text{turn},75,\text{B.A}} = 6.65 + 1.25 = 7.90\,\text{hr} \tag{109}$$

$$t_{\text{melt},70,\text{B.A}} = \frac{371.6 \times 10^6}{13{,}392} = 27.749\,\text{s} = 7.71\,\text{hr} \tag{110}$$

$$t_{\text{turn},70,\text{B.A}} = 7.71 + 1.25 = 8.96\,\text{hr} \tag{111}$$

Case B.B

$$t_{\text{melt},75,\text{B.B}} = \frac{371.6 \times 10^6}{10{,}690} = 34.762\,\text{s} = 9.66\,\text{hr} \tag{112}$$

$$t_{\text{turn},75,\text{B.B}} = 9.66 + 1.25 = 10.91\,\text{hr} \tag{113}$$

$$t_{\text{melt},70,\text{B.B}} = \frac{371.6 \times 10^6}{9{,}182} = 40.471\,\text{s} = 11.24\,\text{hr} \tag{114}$$

$$t_{\text{turn},70,\text{B.B}} = 11.24 + 1.25 = 12.49\,\text{hr} \tag{115}$$

2-tonne processing time summary — all four sub-cases, single T_{10} tank:

Case	Config.	Driver	$t_{\text{melt,2t}}$ (hr)	$t_{\text{turn,2t}}$ (hr)	Slack in 24 hr (hr)
A.A	1.5" sup / 1.5" coil	75 °C	6.34	**7.59**	16.41
A.A	1.5" sup / 1.5" coil	70 °C	7.34	**8.59**	15.41
A.B	1.5" sup / 1" coil	75 °C	9.34	**10.59**	13.41
A.B	1.5" sup / 1" coil	70 °C	10.87	**12.12**	11.88
B.A	1" sup / 1.5" coil	75 °C	6.65	**7.90**	16.10
B.A	1" sup / 1.5" coil	70 °C	7.71	**8.96**	15.04
B.B	1" sup / 1" coil	75 °C	9.66	**10.91**	13.09
B.B	1" sup / 1" coil	70 °C	11.24	**12.49**	11.51

Key findings:

- Every sub-case completes a 2-tonne run well within 24 hours, with a minimum slack of 11.51 hr (Case B.B at 70°C). Even the worst configuration leaves nearly half the day unused on a single-tank basis.
- The fastest single-tank completion is Case A.A at 75 °C: 7.59 hr total, leaving 16.41 hr of spare capacity.
- A.B and B.B (1-inch coil cases) require roughly 3 hr longer than their 1.5-inch coil counterparts — a direct consequence of the 31% smaller coil surface area reducing $\dot{Q}_{\text{net}}$ by ≈5 kW.
- With three T_{10} tanks in staggered operation, the number of 2-tonne runs completable per 24 hours ranges from 5.76 (B.B, 70°C) to 9.49 (A.A, 75°C) — corresponding to 11.5 t/day to 19.0 t/day of net processed output, all comfortably exceeding the 2 t/day mandate.

10 T_5 Replenishment Tank — Sensible Heating

10.1 Process Conditions

Palm oil arrives from T_{10} via HPP at ≈ 36 °C and is trim-heated to the process target of 60 °C. The oil-side capacity rate is:

$$C_{\text{po}} = \frac{2000}{20 \times 3600} \times 2100 = 58.33\,\text{W}\,\text{K}^{-1} \tag{116}$$

In all four sub-cases $C_{\text{po}} \ll C_w$, so $C_r \approx 0$ and $C_{\min} = C_{\text{po}}$ regardless of supply pipe size.

10.2 NTU, Effectiveness, and Oil Outlet Temperature

$$NTU_{\mathrm{T5}} = \frac{U_{\mathrm{liquid}}\,A_{\mathrm{coil,T5}}}{C_{\mathrm{po}}} \qquad \varepsilon_{\mathrm{T5}} = 1 - e^{-NTU_{\mathrm{T5}}} \qquad T_{\mathrm{po,out}} = 36 + \varepsilon_{\mathrm{T5}}\,(75 - 36) \tag{117}$$

Table 9: T_5 sensible heating performance — all four sub-cases.

Case	Coil	$A_{\mathrm{coil,T5}}$ ($\mathrm{m^2}$)	U_{liq} ($\mathrm{W/m^2K}$)	NTU_{T5}	$\varepsilon_{\mathrm{T5}}$	$T_{\mathrm{po,out}}$ (°C)
A.A	1.5-inch	9.88	117.4	19.84	≈1.000	75.0
A.B	1-inch	6.82	118.1	13.81	≈1.000	75.0
B.A	1.5-inch	9.88	115.4	19.50	≈1.000	75.0
B.B	1-inch	6.82	117.1	13.69	≈1.000	75.0

T_5 achieves full thermal equilibration with the hot water supply in all four sub-cases. The extremely high NTU (≥ 13.7 even in the worst case, B.B with the smaller coil) means the palm oil exits at ≈ 75 °C, well above the 60 °C process requirement, for all supply and coil size combinations. A mixing valve or duty-cycle control on the water loop is required in all four configurations to hold the 60 °C outlet setpoint precisely. T_5 performance is entirely insensitive to supply pipe size or coil bore.

11 Process Control Architecture

11.1 Overview

The control system operates on a **demand-driven** logic: T_5 is the process master. A batch cycle in T_{10} initiates *only* when T_5 level or temperature drops below its setpoint. This is fundamentally different from timed batch control.

11.2 Sensor Naming Convention

Table 10: Sensor and signal naming convention used throughout this section.

Symbol	Meaning
θ	All temperature sensors (e.g. θ_S — T_5 temperature, θ_{Sol} — solar loop temperature, θ_H — thermal battery temperature, θ_{10} — T_{10} tank temperature)
α	All float / level sensors (e.g. α_S — T_5 level, α — T_{10} tank level)
$\delta_X, \delta_Y, \delta_Z$	Distribution demand signals from downstream HMIs X, Y, Z

11.3 Derivation of the Dynamic Upper Thermal Boundary (T_f)

To ensure process safety and maintain structural integrity, the system utilizes a dynamic upper thermal boundary constraint ($T_f \leq 55$ °C). While the lower limit (40 °C) is explicitly measured at the tank, the upper threshold fluctuates depending on the thermodynamic capacities of the mixing mediums: water (w), stainless steel structural components (ss), and the process liquid palm oil (p).

The final equilibrium temperature, T_f, is governed by the conservation of energy principle, assuming complete (100%) heat transfer under worst-case thermal absorption conditions:

$$Q_{\text{lost}} = Q_{\text{gained}} \tag{118}$$

The total heat lost by the water and stainless steel component mass cooling from an initial temperature x to T_f is expressed as:

$$Q_{\text{lost}} = (m_w c_w + m_{ss} c_{ss})(x - T_f) \tag{119}$$

Where:

- m_w = mass of water $\approx 90\,\text{kg}$ (derived from $90\,\text{L}$)
- c_w = specific heat capacity of water $\approx 4184\,\text{J/kg}\cdot{}^\circ\text{C}$
- m_{ss} = mass of stainless steel structure $= 205\,\text{kg}$
- c_{ss} = specific heat capacity of stainless steel $\approx 500\,\text{J/kg}\cdot{}^\circ\text{C}$

Substituting these constant nominal values yields:

$$\begin{aligned}
Q_{\text{lost}} &= [(90 \times 4184) + (205 \times 500)]\,(x - T_f) \\
Q_{\text{lost}} &= (376560 + 102500)(x - T_f) \\
Q_{\text{lost}} &= 479060(x - T_f)
\end{aligned} \tag{120}$$

Concurrently, the heat energy gained by the palm oil mass (m_p) shifting from an initial temperature y up to equilibrium T_f is calculated using the standard specific heat capacity of palm oil ($c_p \approx 2000\,\text{J/kg}\cdot{}^\circ\text{C}$):

$$Q_{\text{gained}} = m_p c_p (T_f - y) = 2000 m_p (T_f - y) \tag{121}$$

Equating energy loss to energy gain allows the isolation of the equilibrium parameter:

$$479060(x - T_f) = 2000 m_p (T_f - y) \tag{122}$$

Expanding and solving for the final system equilibrium temperature (T_f) yields the dynamic tracking function:

$$T_f(x, y, m_p) = \frac{479060x + 2000 m_p y}{479060 + 2000 m_p} \tag{123}$$

Dividing by the common scalar factor of 2000 simplifies the expression for efficient implementation within runtime PLC memory registers:

$$T_f(x, y, m_p) = \frac{239.53x + m_p y}{239.53 + m_p} \tag{124}$$

The active safety logic system monitors continuously evaluated inputs (x, y, m_p) and triggers an interlock cycle whenever the function yields $T_f(x, y, m_p) > 55\,{}^\circ\text{C}$.

11.4 Control Loops and Setpoints

Table 11: PLC control loops, sensors, and actions with dynamic thermal boundaries.

Condition	Priority / Mode	Action
$\theta_S < 40\,°\mathrm{C}$	Priority 1	Heat T_5 (open coil loop valve to T_5)
$40\,°\mathrm{C} \leq \theta_S \leq T_f(x, y, m_p)$	—	Normal operating band; no action required
$T_f(x, y, m_p) > 55\,°\mathrm{C}$	Safety / Upper Limit	Dynamic cut-off triggered; throttle process or divert flow
$\alpha_S < 50\%$	Priority 2 (after temp)	Trigger HPP to transfer liquid palm oil, $T_{10} \rightarrow T_5$
$\theta_{\mathrm{Sol}} < 70\,°\mathrm{C}$ **and** $\theta_H > 70\,°\mathrm{C}$	—	Bypass solar loop
$\theta_{\mathrm{Sol}} < 70\,°\mathrm{C}$ **and** $\theta_H < 70\,°\mathrm{C}$	—	Bypass solar loop **and** activate immersion elements
$\theta_H < 60\,°\mathrm{C}$ with elements on	Alarm	Alarm; circulate loop until $\theta_H < 36\,°\mathrm{C}$, then enter thermal recharge
Thermal recharge mode	Alarm	Alarm; cease flow; apply full auxiliary heat output; resume heating the highest-ranked T_{10} tank when $\theta_{\mathrm{Sol}} > 70\,°\mathrm{C}$
T_{10} surplus (T_5 not in demand)	—	Route hot water to T_{10} until $\alpha_S > 90\%$
T_{10} high-limit	Safety	Close coil valve; alarm if $T_{10} > 75\,°\mathrm{C}$

Note on thermal recharge: Recharge mode is entered only after the $\theta_H < 60\,°\mathrm{C}$ alarm condition persists and the battery has been allowed to circulate down to $\theta_H < 36\,°\mathrm{C}$. During recharge, flow to all downstream loops ceases and the full auxiliary heating capacity is dedicated to restoring θ_H. If solar availability returns during recharge ($\theta_{\mathrm{Sol}} > 70\,°\mathrm{C}$), the PLC may redirect heating priority to the highest-ranked T_{10} tank to make best use of the available thermal input.

11.5 Tank Selection Scoring System

Because three T_{10} tanks (A, B, C) compete for both water-heating priority and oil draw-off priority, each tank is continuously assigned two scores by the PLC:

$$S_{\mathrm{casual}} = 0.6\,\alpha + 0.4\,\theta \quad \text{(volume-weighted; used for normal draw)} \tag{125}$$

$$S_{\mathrm{immediate}} = 0.4\,\alpha + 0.6\,\theta \quad \text{(temperature-weighted; used for urgent draw)} \tag{126}$$

where α and θ are the normalised level and temperature readings of the individual T_{10} tank in question.

- Under normal operating conditions, the PLC ranks $T_{10,A}$, $T_{10,B}$, and $T_{10,C}$ by S_{casual}; the highest-scored tank is drawn from first.
- Under urgent/low-buffer conditions, ranking switches to $S_{\mathrm{immediate}}$, weighting temperature more heavily so that the nearest-to-target tank is prioritised.
- Water-heating priority follows the *same* ranking: the highest-scored tank (by the applicable score) also receives the next allocation of hot water from the thermal battery.

11.6 Oil Distribution Logic

T_5 supplies liquid palm oil to three downstream process lines (X, Y, Z) through three proportional control valves. Each line reports a continuous demand signal (δ_X, δ_Y, δ_Z) from its respective HMI.

Table 12: T_5 outlet distribution logic.

Condition	Action
$\delta_X + \delta_Y + \delta_Z \leq \mathrm{Supply}$	All three valves open proportionally; free flow to all lines
$\delta_X + \delta_Y + \delta_Z > \mathrm{Supply}$	Priority queue applies: line X is served to its full demand first, then Y, then Z, until supply is exhausted

- The X $\rightarrow$ Y $\rightarrow$ Z priority order is configurable by the PLC programmer via the HMI; it is **not** hardwired and may be re-ordered to match changing production priorities.
- All three outlet valves are proportional (modulating) control valves, not simple on/off devices, allowing partial opening to match any demand level below full flow.

11.7 Priority Logic and Solar Circulation Modes

The PLC operates on the following priority hierarchy:

1. **T_5 temperature** (θ_S) is the primary priority. If $\theta_S < 40\,°C$, T_5's coil loop is active.
2. When T_5 is satisfied, the water loop is directed to whichever T_{10} tank ranks highest by the applicable scoring system (Section 11.5).
3. **Three solar circulation modes govern loop routing:**
   - *Through-solar*: Loop water passes through the solar collector array, drawing active heat from the collectors into the system. This mode is used when the collectors are producing useful heat ($\theta_{Sol} \geq 70\,°C$) and the reservoir is below target.
   - *Bypass-solar*: The solar collector array is bypassed entirely. This mode is engaged when $\theta_{Sol} < 70\,°C$ (collectors not generating useful heat) or when the reservoir is already at or above T_{hot}.
   - *Quick-solar cycle*: A short gravity-assisted circulation stroke used to prime the solar loop pump at start-up, confirming flow before committing to through-solar operation.
4. **Thermal battery interlock**: If $\theta_H < 60\,°C$ with immersion elements already energised, the alarm and recharge sequence of Section 11.3 governs.

11.8 Batch Release Condition

A T_{10} batch is *only released* (HPP activated to T_5) when:

$$\alpha_S < 50\% \quad \textbf{AND} \quad \theta_{10}\big(\text{highest-ranked tank}\big) \geq 36\,°C \tag{127}$$

The tank selected to supply the batch is the one ranked highest by $S_{\text{immediate}}$ at the moment of release. This demand-pull logic prevents premature extraction of unmelted oil and ensures T_5 never starves the process lines.

12 Design Validation Summary

Table 13: Design parameter summary (1 of 2) — vessel, coil, and thermal performance across all four supply/coil sub-cases. Prefix letter: supply pipe (A = 1.5-inch, B = 1-inch). Suffix letter: coil bore (A = 1.5-inch, B = 1-inch).

Parameter	A.A	A.B	B.A	B.B
Vessel — T_{10} (common to all cases)				
External surface area, A_{vessel} (m^2)	20.916			
Tank heat loss, melt phase at 36°C (W)	1072			
Coil — T_{10}				
Coil material	SS 304 Sch 10S			
Coil NPS (inch)	1.5	1	1.5	1
Practical coil length, L_{prac} (m)	90.0			
Coil surface area, A_{coil} (m^2)	13.68	9.44	13.68	9.44
Water-side h_i ($W/m^2{\cdot}K$)	9,147	19,594	4,166	8,897
U_{solid} ($W/m^2{\cdot}K$)†	33.3–33.5 (effectively constant)			
U_{liquid} ($W/m^2{\cdot}K$)	117.4	118.1	115.4	117.1
NTU-ε — Melting Phase (solid, 75°C drive)				
NTU_{solid}	0.0520	0.0360	0.1329	0.0924
ε_{solid}	0.0506	0.0354	0.1245	0.0882
$\dot{Q}_{net}$ at 75°C (kW)	16.28	11.05	15.52	10.69
$\dot{Q}_{net}$ at 70°C (kW)	14.05	9.50	13.39	9.18
Energy Budget — Full 10 kL Batch (common to all cases)				
$E_{sensible}$, 18.1→36°C (MJ)	318.6			
E_{latent}, fusion (MJ)	1335.0			
E_{total} (MJ)	1653.6			
Melting rate at 75°C (kg/hr)	315.4	214.1	300.7	207.1

† U_{solid} ranges from 33.29 to 33.51 $W\,m^{-2}\,K^{-1}$ across all four cases (<0.7% spread). The outer oil-side resistance ($1/h_{o,solid}$) accounts for >95% of $1/U_{solid}$ in every configuration, rendering U_{solid} insensitive to both supply pipe size and coil bore.

Table 14: Design parameter summary (2 of 2) — production capacity, solar integration, and T_5 replenishment tank across all four sub-cases.

Parameter	A.A	A.B	B.A	B.B
Full-Batch Production Capacity (single T_{10})				
Melt time at 75°C (hr)	28.22	41.56	29.60	42.97
Melt time at 70°C (hr)	32.68	48.36	34.30	50.03
24-hr capacity at 75°C (t/day)	7.25	4.99	6.92	4.83
24-hr capacity at 70°C (t/day)	6.30	4.31	6.01	4.17
Safety margin vs 2 t (at 75°C)	3.63×	2.50×	3.46×	2.42×
2-Tonne Target Processing Time (single T_{10}, 75°C drive)				
Melt time for 2 t (hr)	6.34	9.34	6.65	9.66
Total turnaround for 2 t (hr)	7.59	10.59	7.90	10.91
Slack remaining in 24 hr (hr)	16.41	13.41	16.10	13.09
Solar Integration (common to all cases)				
FPC solar yield, E_{solar} (MJ)	118.45			
ETC solar yield, E_{solar} (MJ)	228.40			
FPC coverage — full batch (%)	7.16			
ETC coverage — full batch (%)	13.81			
FPC coverage — 2-tonne target (%)	31.88			
ETC coverage — 2-tonne target (%)	**61.46**			
Auxiliary energy, ETC, 2-t target (MJ)	143.20			
T_5 Replenishment Tank				
Coil surface area, $A_{\text{coil,T5}}$ (m^2)	9.88	6.82	9.88	6.82
NTU_5	19.84	13.81	19.50	13.69
ε_5	≈1.000 (all cases)			
Palm oil outlet temperature (°C)	40.0 - 55.0			
Process target (°C)	55.0			
Thermal margin (°C)	20.0			

13 Conclusions and Recommendations

1. **2-tonne target is met with large margin.** Even in the worst-case 70°C driver scenario with verified uninsulated tank losses, a single T_{10} tank delivers 4.17 t/day — a 2.42× safety factor.

2. **The bottleneck is the solid-phase U.** At $U_{\text{solid}} = 33.3\,\mathrm{W\,m^{-2}\,K^{-1}}$, melting takes 42.97 hr per full tank. If future production targets rise significantly, options include: (a) mechanical agitation as soon as a liquid layer forms (immediately boosts h_o to $\geq 100\,\mathrm{W\,m^{-2}\,K^{-1}}$), or (b) raising driver temperature to 80 °C.

3. **Tank insulation is recommended.** At-melt-phase losses of 1.07 kW grow to $\approx$ 2.6 kW at hold temperature. Insulating T_{10} with 50 mm mineral wool would reduce losses by $\approx$ 85% and reduce melt time by approximately 1.5 hr.

4. **Thermal recharge interlock is critical.** The $\theta_H < 60\,°\mathrm{C}$ alarm and recharge sequence must be hardwired as a fail-safe, not software-only, to protect the pump from running against a hot but thermally depleted loop.

5. **ETC is the preferred collector technology; a break-even analysis is recommended before final procurement.** On thermal performance alone, the evacuated tube collector (ETC) configuration delivers **61.46%** solar coverage of the 2 t/day production energy demand versus **31.87%** for the flat plate collector (FPC) — a **29.6 percentage point** improvement, or approximately 1.93× the solar yield for the same collector aperture area. Correspondingly, auxiliary energy consumption is reduced by 109.95 MJ per day under ETC operation. However, thermal performance figures alone do not constitute a procurement decision. ETC units carry a higher capital cost per unit aperture area, and whether that premium is justified depends on the ratio of installation cost differential to the cumulative auxiliary energy savings over the system's service life. If, for instance, the ETC array commands only a modest cost premium relative to an equivalent FPC installation, the case for ETC becomes compelling given the near-doubling in solar coverage. Conversely, if the FPC option is substantially cheaper but delivers less than half the solar offset, the auxiliary energy cost penalty over multi-year operation is likely to erode and eventually reverse any initial capital saving. It is therefore strongly recommended that a formal break-even and lifecycle cost analysis be conducted — incorporating collector procurement and installation costs, auxiliary energy tariffs, expected degradation rates, and maintenance schedules — prior to final specification. The thermal analysis presented in this document provides the performance baseline from which that financial assessment should proceed.

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